KATULA VAISHNAVI

Hyderabad, Telangana | kvaishu0405@gmail.com | +91 7416842198 | LinkedIn | GitHub

SUMMARY

B.Tech Data Science student (2027) with hands-on experience building end-to-end ML systems spanning supervised learning, LLM fine-tuning, retrieval-augmented generation, and multi-agent AI. Strong foundation in model evaluation, optimization, and production-grade ML pipelines, with a focus on rigorous experimentation and measurable results.

EDUCATION

Malla Reddy Deemed to be University — Hyderabad, Telangana

B.Tech in Data Science (CGPA: 7.81) — *Aug 2023 – Jun 2027*

EXPERIENCE

Edunet Foundation — Hyderabad

AI and Cloud Intern — *Jun 2025 – Aug 2025*

- Built a multi-model Disease Prediction system on a clinical dataset, benchmarking Logistic Regression, Random Forest, and XGBoost classifiers for diagnostic decision support.
- Performed feature engineering, handling of missing/imbalanced data, and hyperparameter tuning, improving prediction accuracy and recall on minority disease classes.
- Evaluated models using precision, recall, F1-score, and ROC-AUC, and presented results with SHAP-based explainability to highlight key clinical risk factors.
- Deployed the trained model via a lightweight Flask API and documented the pipeline using AWS cloud fundamentals for future scalable deployment.

TECHNICAL SKILLS

Languages: Python, SQL

ML & AI: Scikit-learn, PyTorch, TensorFlow, XGBoost, LightGBM, SHAP, HuggingFace Transformers, PEFT/QLoRA, DistilBERT, LSTM, SVM, RAGAS, Prompt Engineering, Generative AI (LLMs)

Data, DB & Tools: Pandas, NumPy, FAISS, ChromaDB, LangChain, LangGraph, NoSQL, ETL, Power BI, Matplotlib, Seaborn

Engineering: FastAPI, Flask, Cloud (Basics), Streamlit, React, Docker, Git, AWS, Azure, GCP

PROJECTS

AmazonQA-RAG | Python, QLoRA, FAISS, LangChain, RAGAS — GitHub

- Fine-tuned an LLM via QLoRA (4-bit NF4 + LoRA adapters) for efficient, low-resource instruction adaptation on Amazon Q&A data.
- Built a FAISS-backed RAG pipeline with sentence-transformer embeddings, benchmarking 3 variants (Base, RAG, Finetuned) on RAGAS faithfulness/relevancy scores, deployed via Streamlit.

Retail Sales Data Warehouse | Python, PostgreSQL, Airflow, dbt, Docker, Power BI — GitHub

- Built a star-schema DWH (SCD Type 2, 2 fact tables) processing 10K rows / $2.3M revenue through an Airflow DAG gated by 10 data quality checks.
- Delivered 3 business marts (customer-360 RFM, inventory turnover, daily sales KPIs), cutting reporting time from 4 hrs to 8 min.

Matrix Factorisation Recommender | Python, PyTorch, SVD, NCF, BPR — GitHub

- Implemented and benchmarked 5 models (SVD, NCF, BPR vs. Popularity, Item-kNN) using a 1-vs-99 protocol on NDCG@10/MAP@10.
- Designed a cold-start evaluation subset and ran hyperparameter ablations across embedding dimension, negative sampling ratio, and dropout, serving top models via FastAPI.

CERTIFICATIONS

- Google GenAI AI/ML – Hack2Skill, 2025
- Introduction to Technology and AWS – Coursera, 2025
- Microsoft Azure AI-900 – Certiport / PearsonVUE, 2023
- Infosys Pragati Cohort – SpringBoard, 2025
- AWS Learning: Generative AI – AWS Skill Builder, 2026